Weeks 11-13

UMak-CCIS
2026-2027

Basic Router Management

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Expected Learning Outcome:

1. Explain what is Router Management
2. Discuss Router Interface
3. Enumerate Internal and external connections
4. Demonstrate simple router configuration

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What is Router Management?

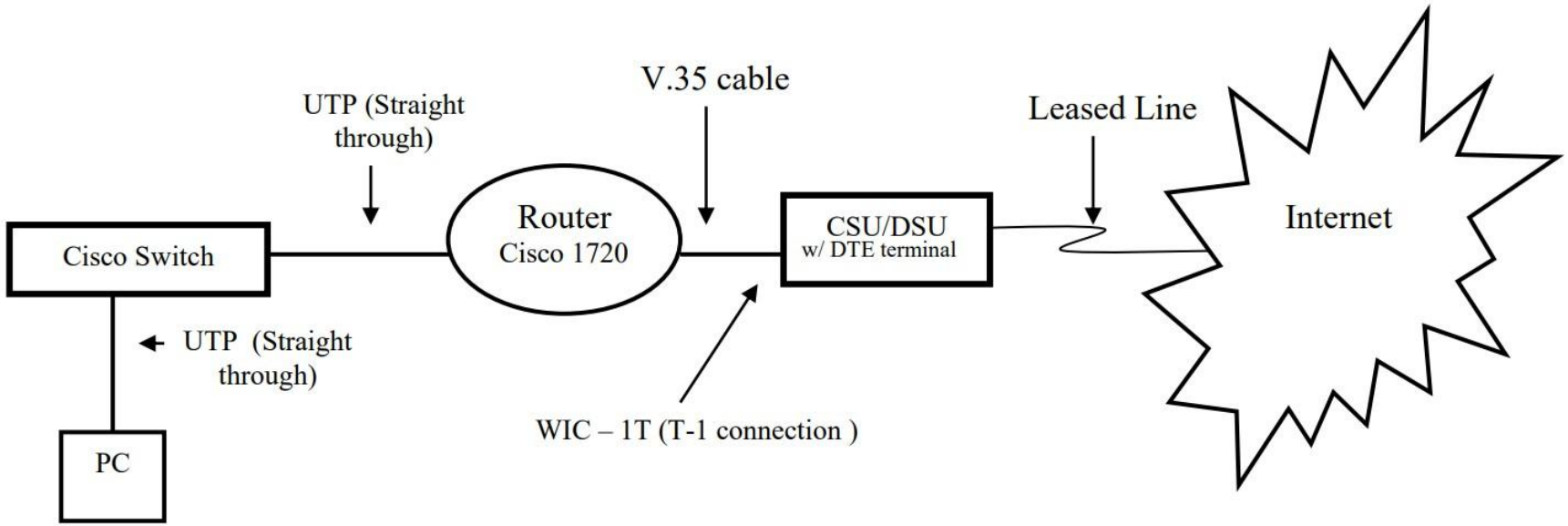
Router management is the process of configuring and monitoring the different routers in your network. If you run a large business, you likely have multiple assets connected to various devices. It can be challenging to keep an eye on every router.

Router management help you keep track of these hardware assets, monitor their status, and check their performance. You'll have access to a full inventory of routers and the devices connected to them. As such, you can troubleshoot network issues and identify faulty hardware before it starts causing serious problems.

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Basic Router Interface

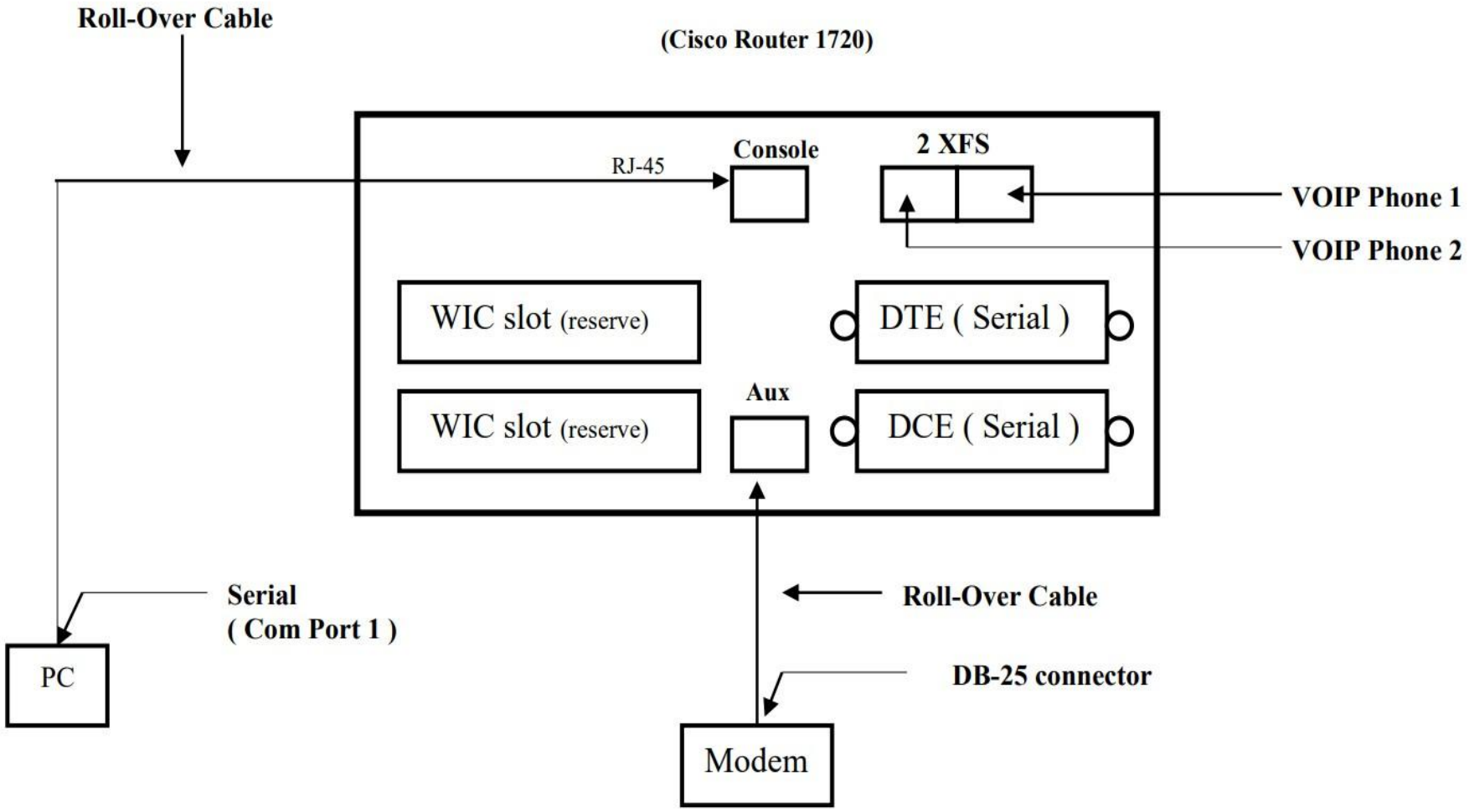
Physical Configuration



CSU – Channel Service Unit
DSU – Data Service Unit
DTE - Data Terminal Equipment

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External Interface



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Internal Configuration

1. After physical configuration and switch on the network
2. Open Hyper Terminal
3. It begin Initialized
4. After Initialization, you will see " Would you like to enter basic management setup?"
5. Press CTRL C
6. Router> will displays it is called "setup mode"
7. Type ENABLE and press enter
8. Router# will displays, it is called "Privilege Mode"
9. You can do the following under Privilege Mode
    a. show ip interface brief
    b. copy run start
    c. show run
    d. show flash
    e. show version
    f. show interface S0
    g. show cdp neighbors
10. To change from privilege mode to configuration mode type configuration terminal or "conf t".
11. Router(config)# will displays

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Line configuration mode

Line configuration mode contents of the following:
a-1. Line console 0
a-2. Line VTY 0 4
a-3. Line Aux 0

Sample Commands:

```
Val# conf t
Val(config)# enable secret pass
Val(config)# line console 0
Val(config)# password pass
Val(config)# login
Val(config)# exec-timeout 0 0
Val(config)# logging synchronous
Val(config)# ^Z
```



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Line configuration mode

```
Val# conf t
Val(config)# line VTY 0 4
Val(config)# password pass2
Val(config)# login
Val(config)# exec-timeout 0 0
Val(config)# ^Z
```



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Line configuration mode

Val# conf t
Val(config)# line aux 0
Val(config)# password **pass3**
Val(config)# login
Val(config)# ^Z
Val#
To encrypt password, do the following

Val# conf t
Val(config)# service password-encryption
Val(config)# ^Z
Val#copy run start

To view your work do the following
Val#show start



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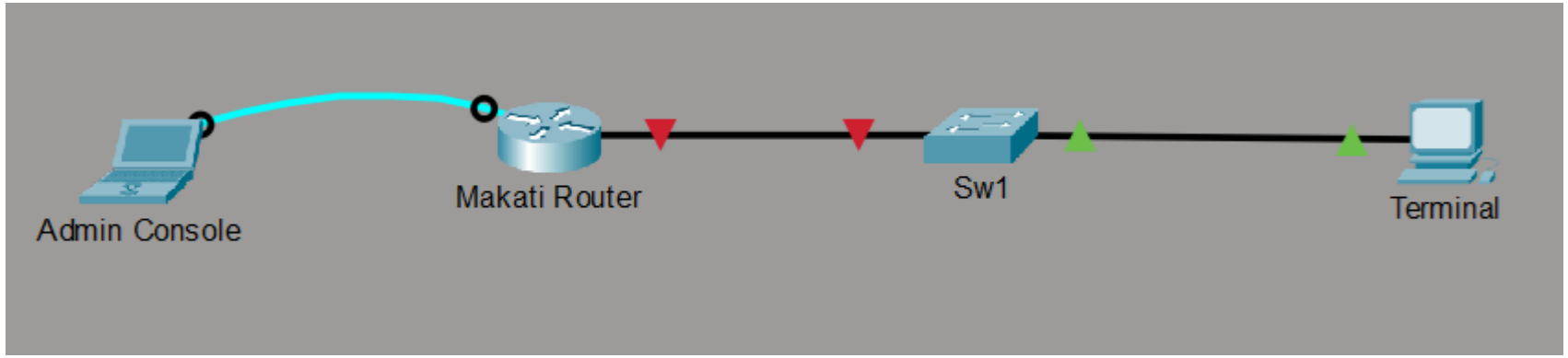
Sample interface configuration

```
Val>en
Val#conf t.
Val(config)#int f0/0
Val(config-if)#ip add 192.168.1.1 255.255.255.0
Val(config-if)#no shut
Val(config-if)#exit
```



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Sample router configuration (demo..)



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Introduction to Network Simulators

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Network Simulator

It is a software tool develops to support computer **network**s analysis and design. Some of the **network simulators** are of general-purpose use and other dedicated to simulate particular types of computer **network**s.

Network Simulation

In computer network research, network simulation is a technique whereby a software program replicates the behavior of a real network. This is achieved by calculating the interactions between the different network entities such as routers, switches, nodes, access points, links, etc



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Network Simulators

BOSON NETSIM https://www.boson.com/netsim-cisco-network-simulator



The Boson™ NetSim™ Network Simulator™ is an application that simulates Cisco Systems' networking hardware and software and is designed to aid the user in learning the Cisco IOS command structure. With NetSim you can learn and master the skills necessary to successfully complete your Cisco certification.

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Network Simulators



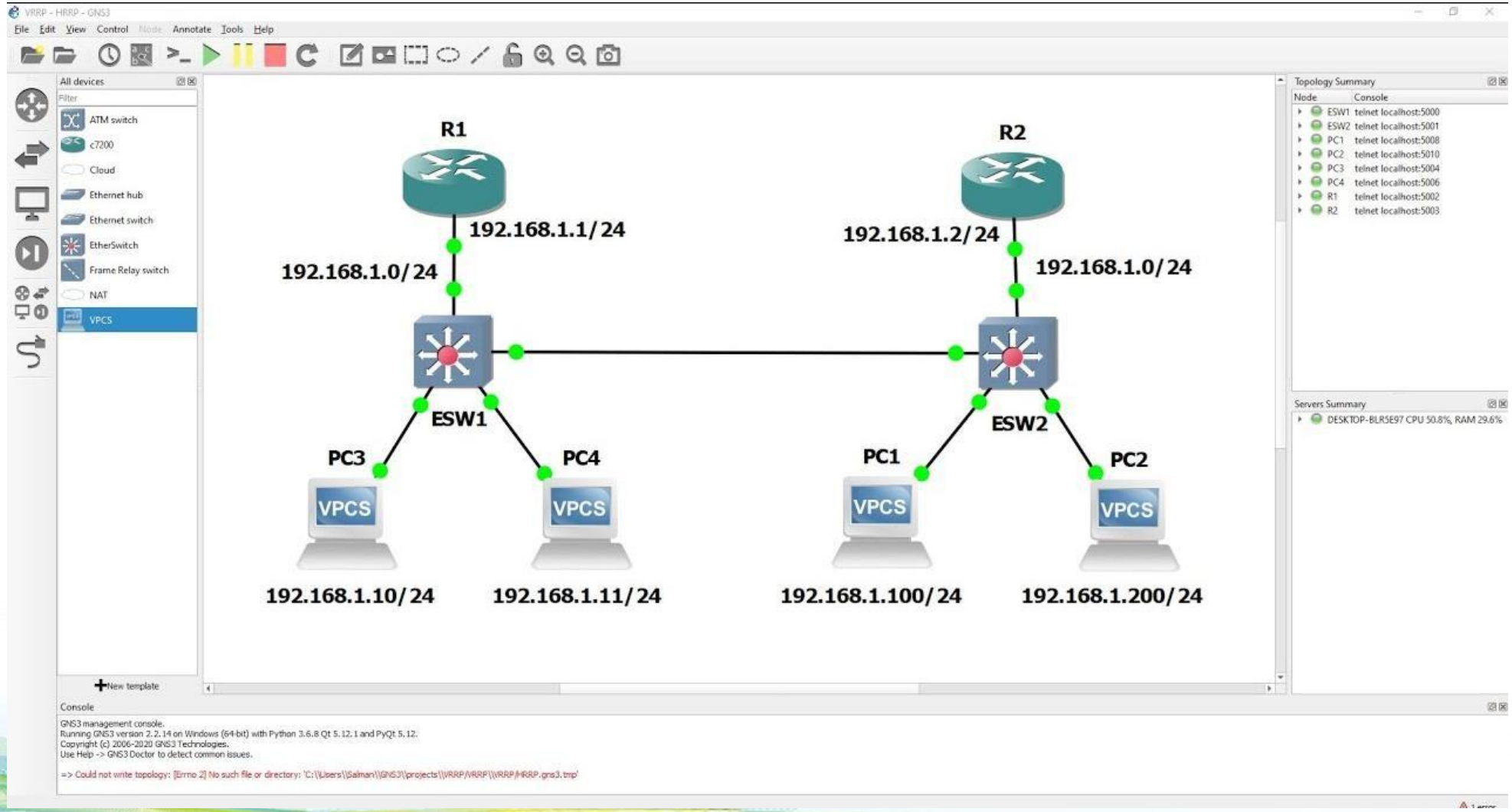
Graphical Network Simulator-3
GNS-3

https://www.gns3.com/

is a network software emulator first released in 2008. It allows the combination of virtual and real devices, used to simulate complex networks. It uses Dynamips emulation software to simulate Cisco IOS.

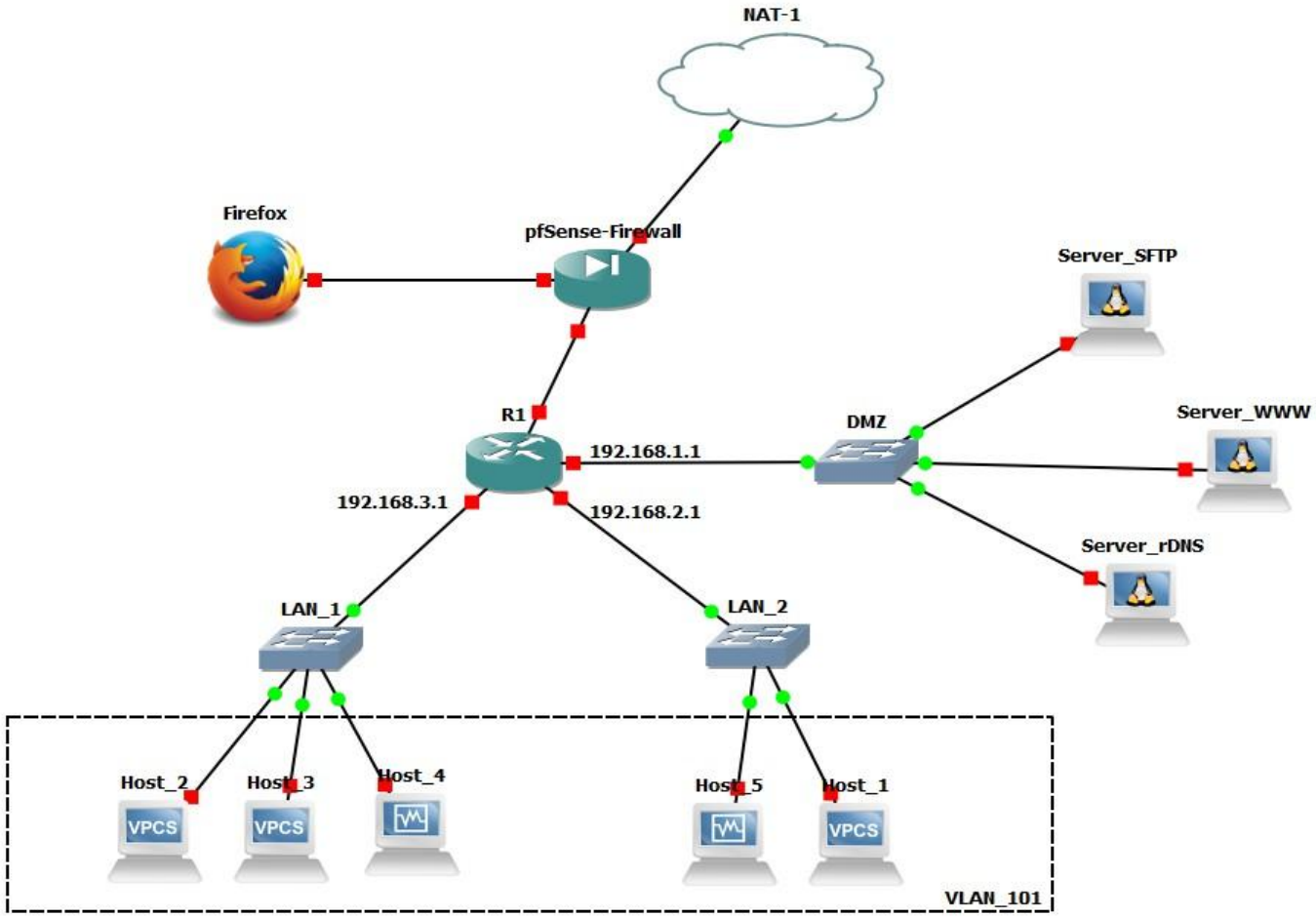
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GNS3 ENVIRONMENT



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GNS3 ENVIRONMENT



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Cisco Packet Tracer

https://www.netacad.com/courses/packet-tracer

Packet Tracer is a cross-platform visual simulation tool designed by Cisco Systems that allows users to create network topologies and imitate modern computer networks. The software allows users to simulate the configuration of Cisco routers and switches using a simulated command line interface.

Packet Tracer supplements physical equipment in the classroom by allowing students to create a network with an almost unlimited number of devices, encouraging practice, discovery, and troubleshooting. The simulation-based learning environment helps students develop 21st century skills such as decision making, creative and critical thinking, and problem solving.

Platform, Microsoft Windows, Android, iOS, macOS

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Cisco Packet Tracer : Download and Installation

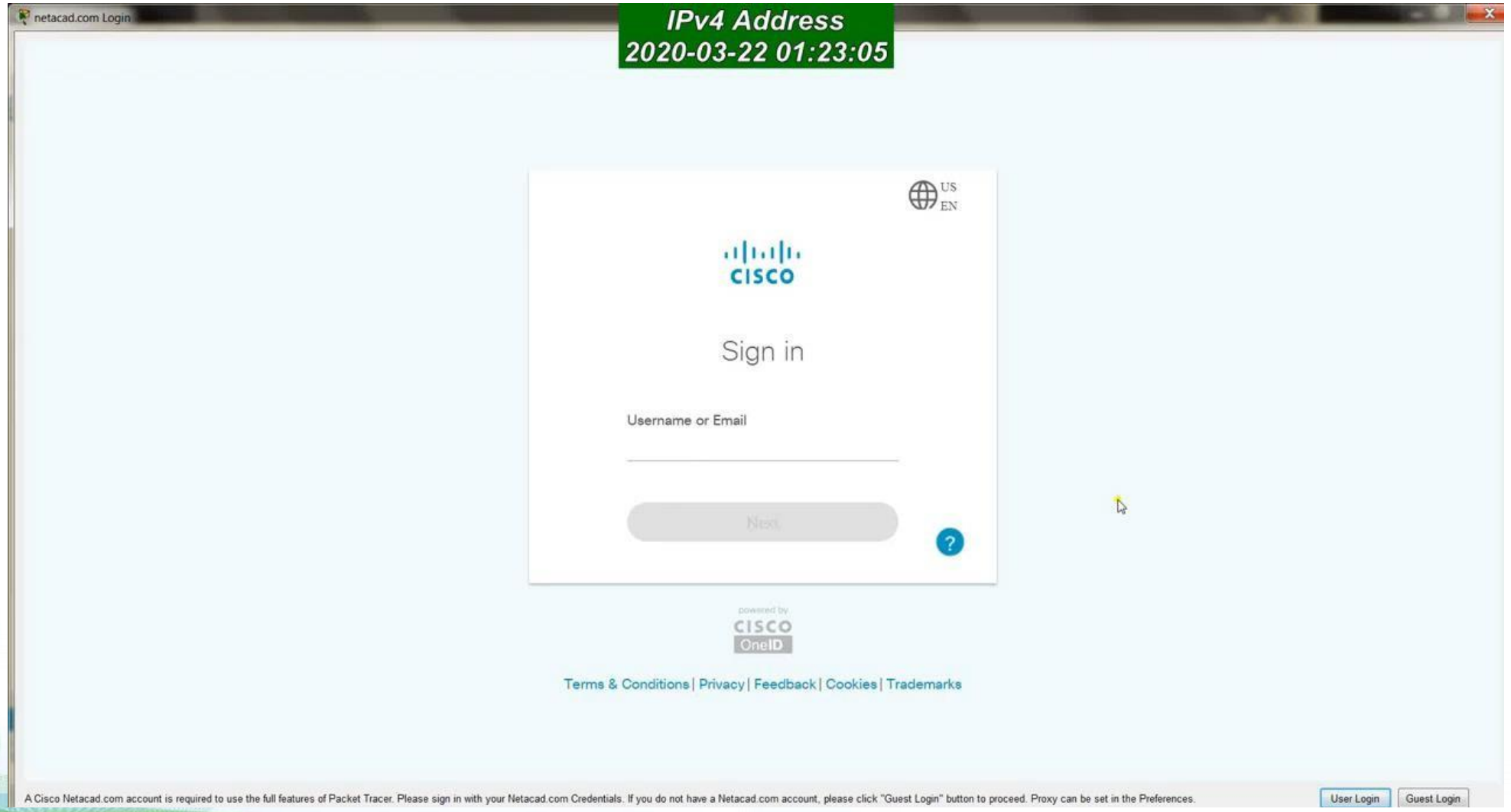
Networking Academy



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Cisco Packet Tracer : Explore Features and functions



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Target Activity:

Basic Router Configuration

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Thank You

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